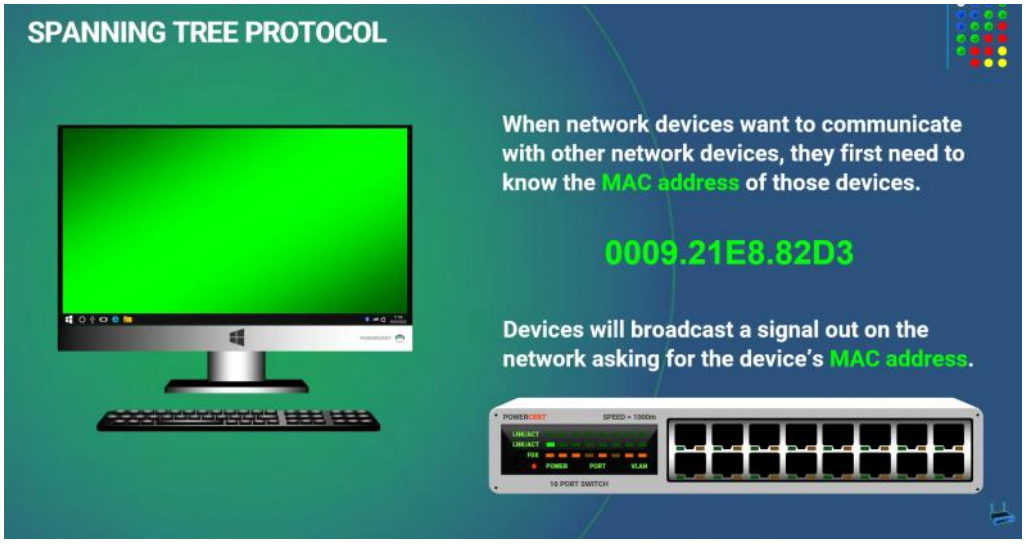
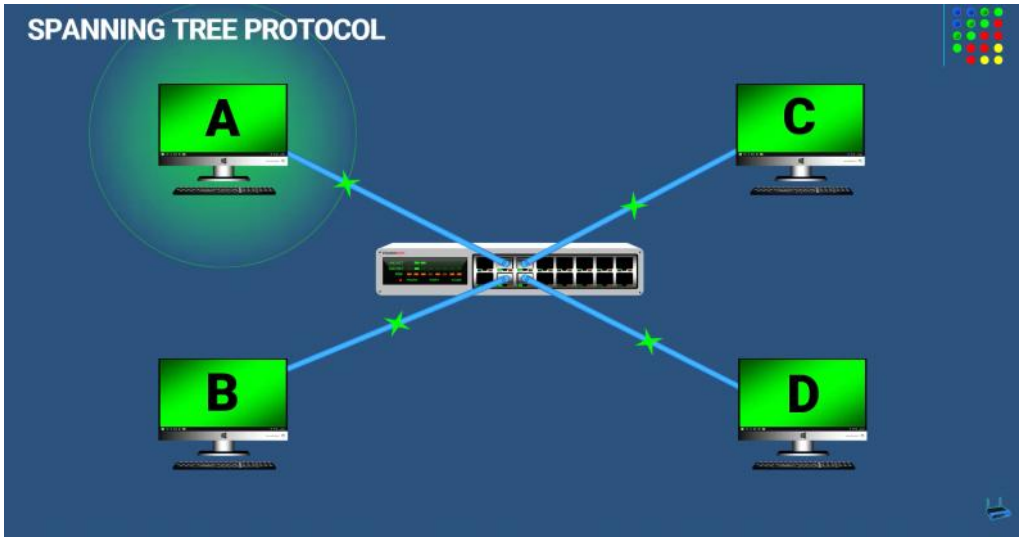
Spanning Tree Protocol

Friday, April 17, 2026 5:33 PM

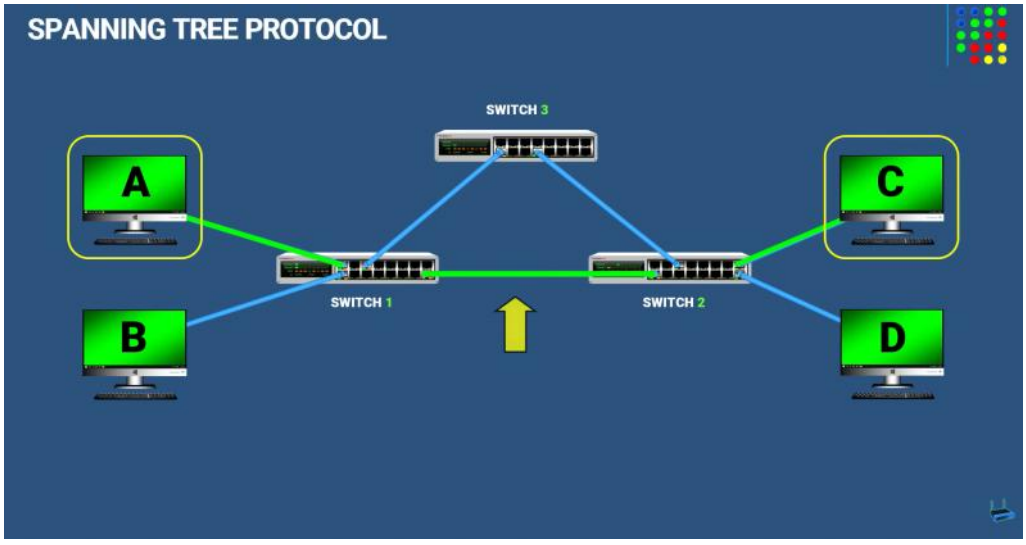
Spanning Tree Protocol (STP)



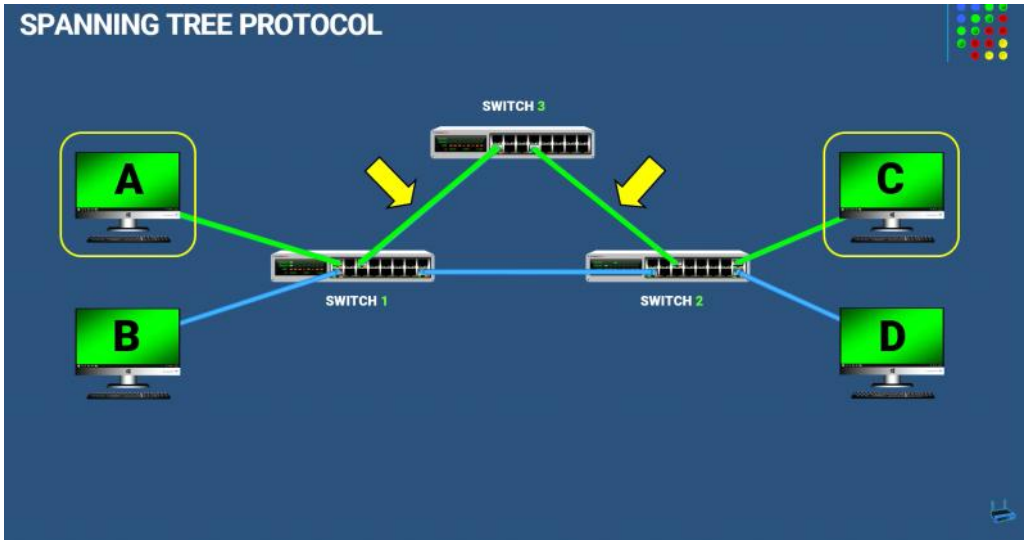


Here when computer A wants to communicate with any other computer on this network then, computer A sends a data packet to swi tch which is called broadcast frame and once it receives the switch then it forwards it to every device which is connected to it.

But in some cases for redundancies, network administrator may add more switches which is show below -



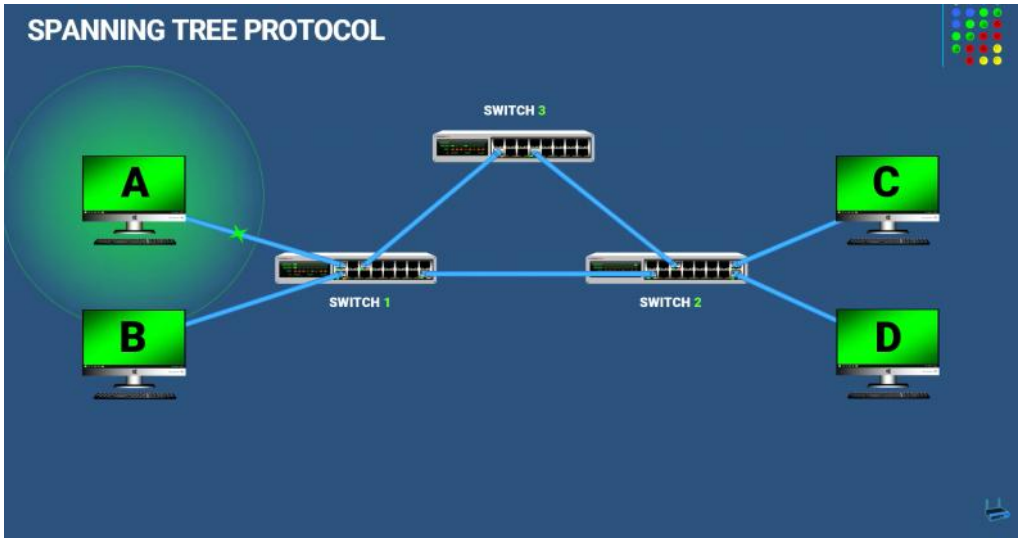
Here as you can see, if computer A wants to communicate with computer C then it may choose this path or if this path gets dow n then it can choose this path -



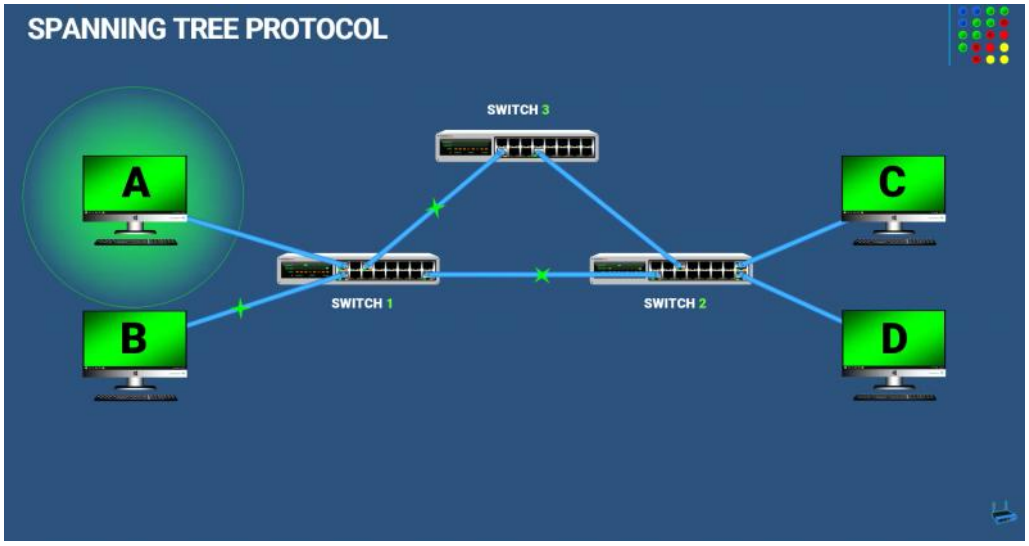
But the problem that may occur while making these kinds of topology is that it may create broadcast loops -

How does his occur? It is shown below -

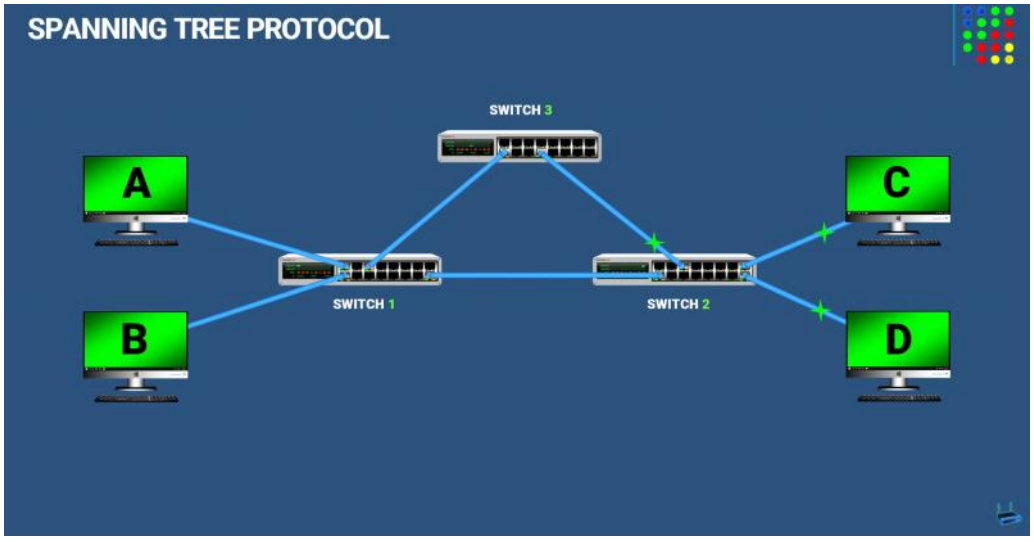
When computer A wants to communicate with computer C then it will send a broadcast frame to switch 1 as shown below -



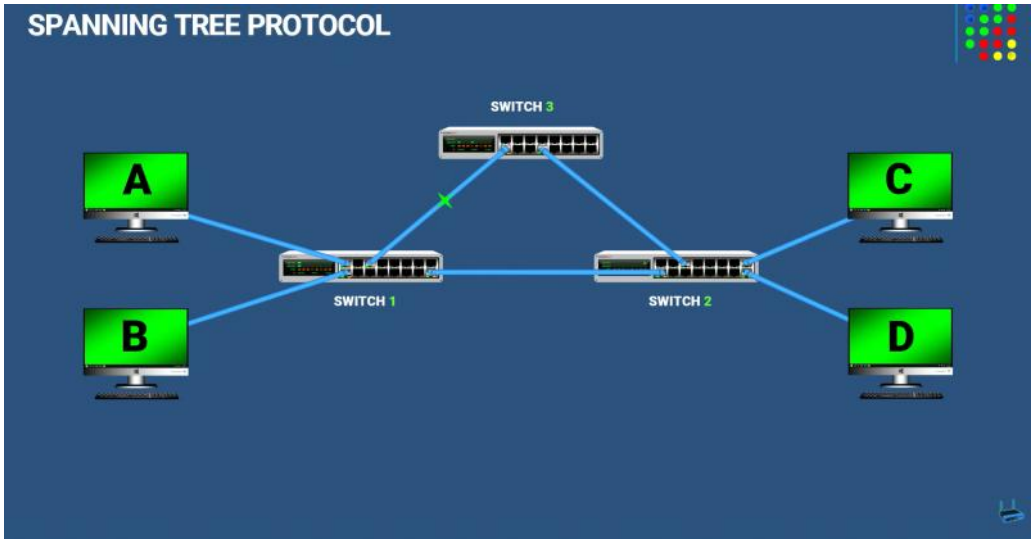
But when switch 1 receives it, it will send the data to all the device connected on it like switch 3 and 4 as well as compute r B as shown below -



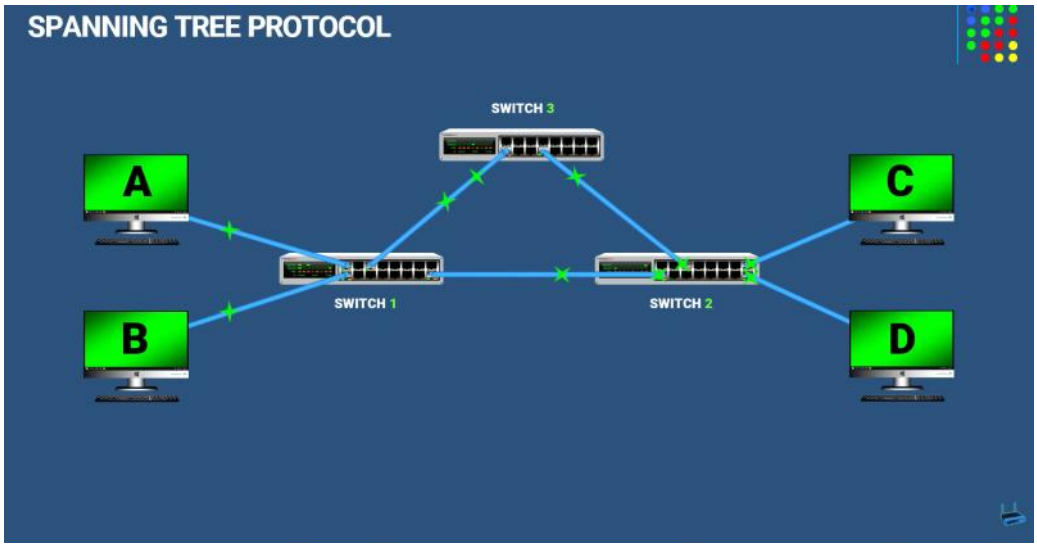
And when switch 2 receives the broadcast, it will forward to computer C and D as well as to switch 3 as shown below -



And when switch 3 receives the broadcast, it will send to switch 1



And when switch 1 receives the broadcast, it will send to switch 2 and computer B. which will create a continuous never endin g loop called broadcast storm which will make the network go down.

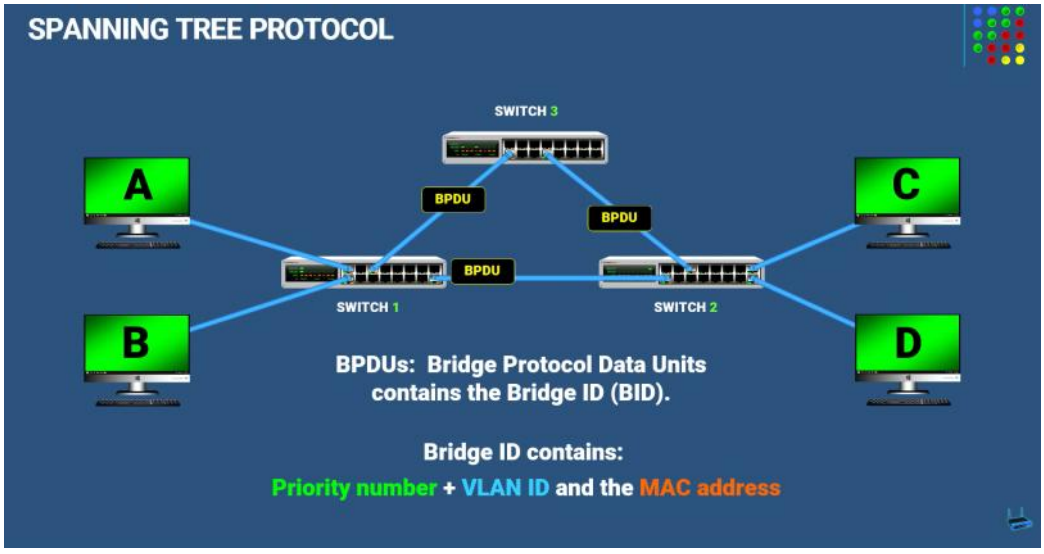


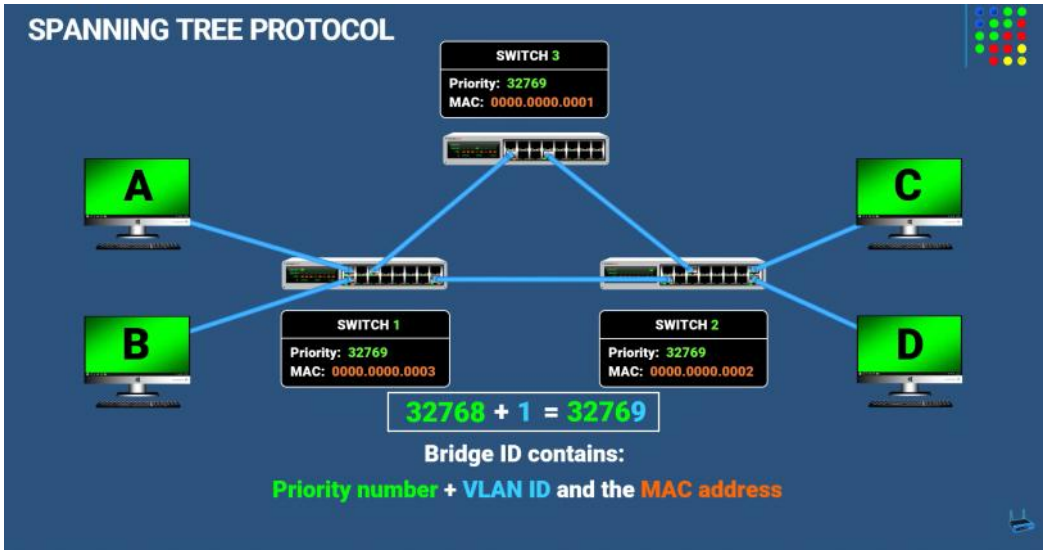
Broadcast Loops and Storms: When multiple switches are connected redundantly to ensure network reliability, broadcast frames can circulate endlessly. This creates a broadcast loop, which consumes all available network bandwidth, leads to a broadcast storm, and brings the entire network to a standsti ll.

This is why STP was invented.

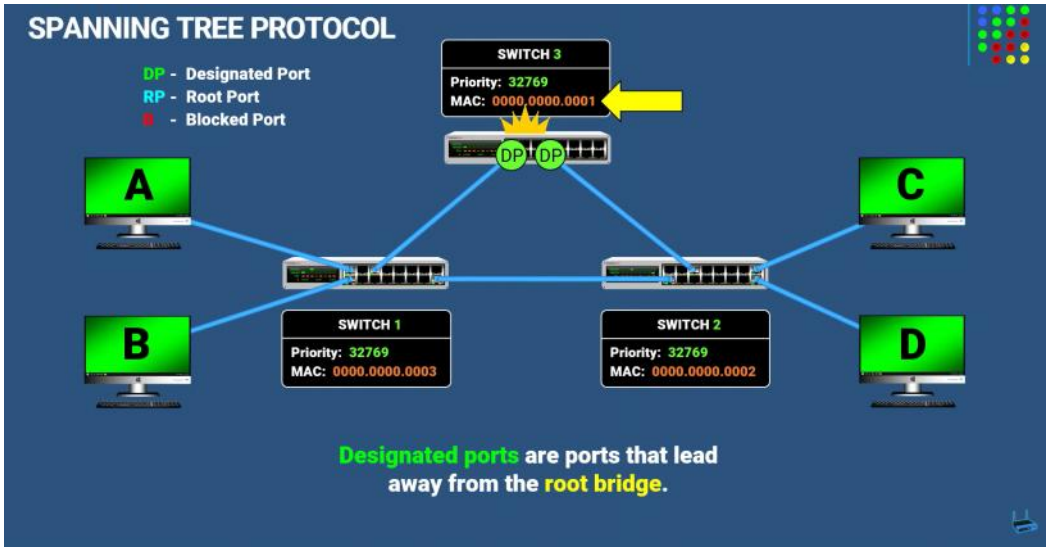
How STP Prevents Loops: The Spanning Tree Protocol solves this by selectively blocking redundant ports to ensure there is onl y one active path between any two devices. The process involves:

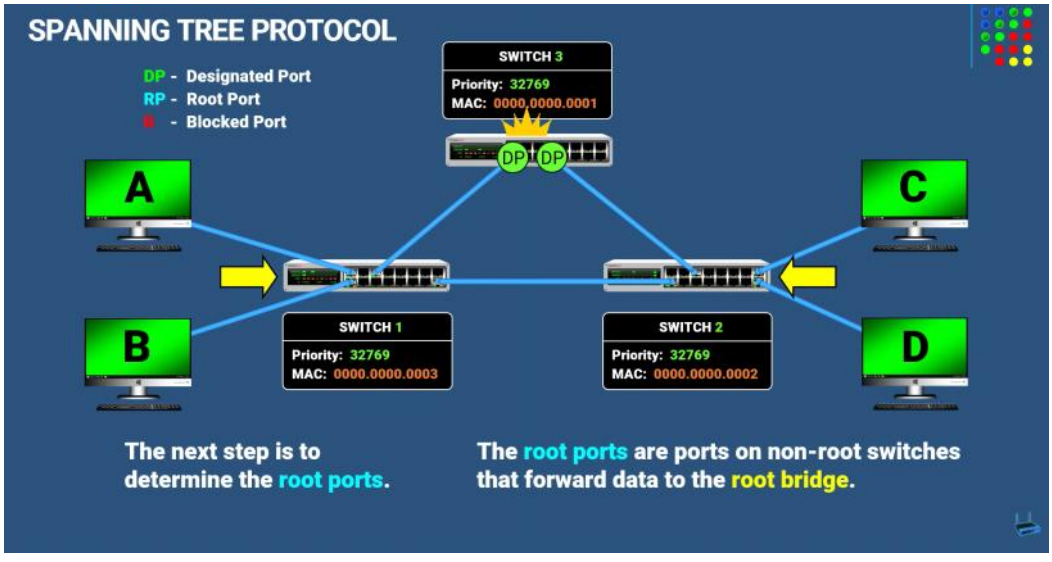
Root Bridge Election: Switches communicate using Bridge Protocol Data Units (BPDUs) to elect a "root bridge" based on the low est Bridge ID (Priority + MAC address).

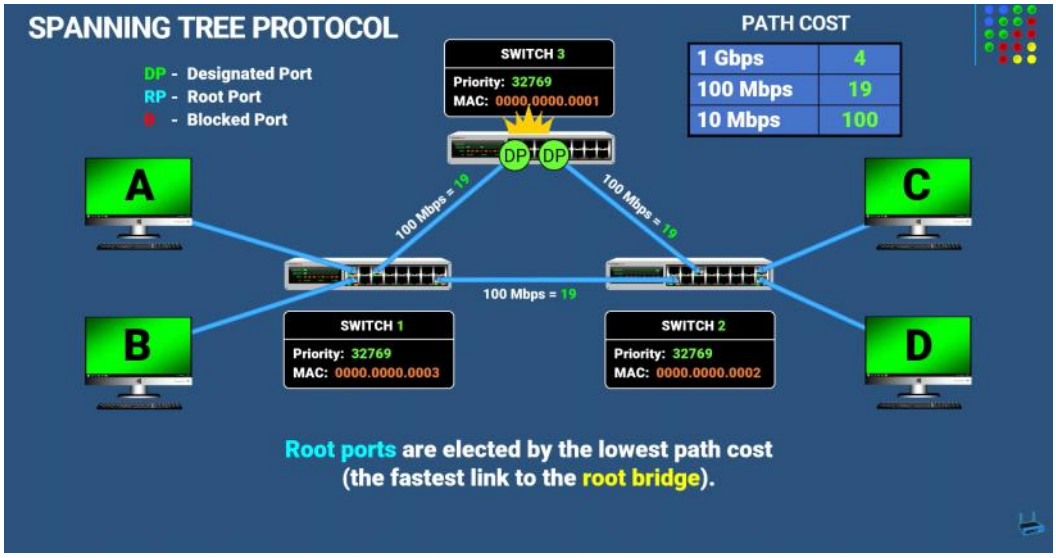


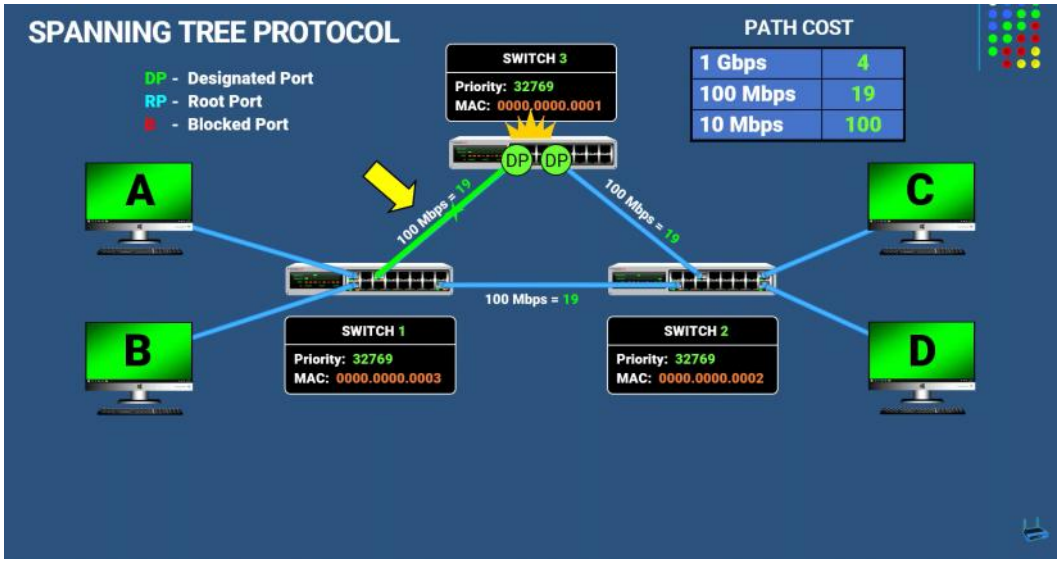


Root Port Selection: Non-root switches determine their "root port" by identifying the path with the lowest cumulative path cost back to the root bridg e.






SPANNING TREE PROTOCOL
DP - Designated Port
RP - Root Port
B - Blocked Port
PATH COST
1 Gbps 4
100 Mbps 19
10 Mbps 100
SWITCH 3
Priority: 32769
MAC: 0000.0000.0001
DP
DP
100 Mbps = 19
100 Mbps = 19
100 Mbps = 19
A
B
C
D
SWITCH 1
Priority: 32769
MAC: 0000.0000.0003
SWITCH 2
Priority: 32769
MAC: 0000.0000.0002
Root ports are elected by the lowest path cost
(the fastest link to the root bridge).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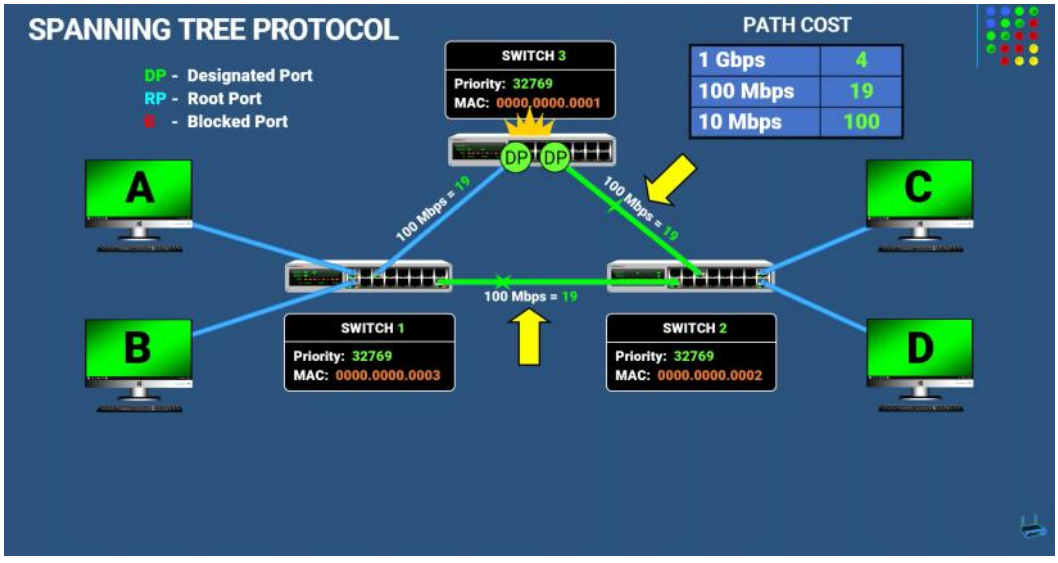
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RP - Root Port
B - Blocked Port
PATH COST
1 Gbps 4
100 Mbps 19
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SWITCH 3
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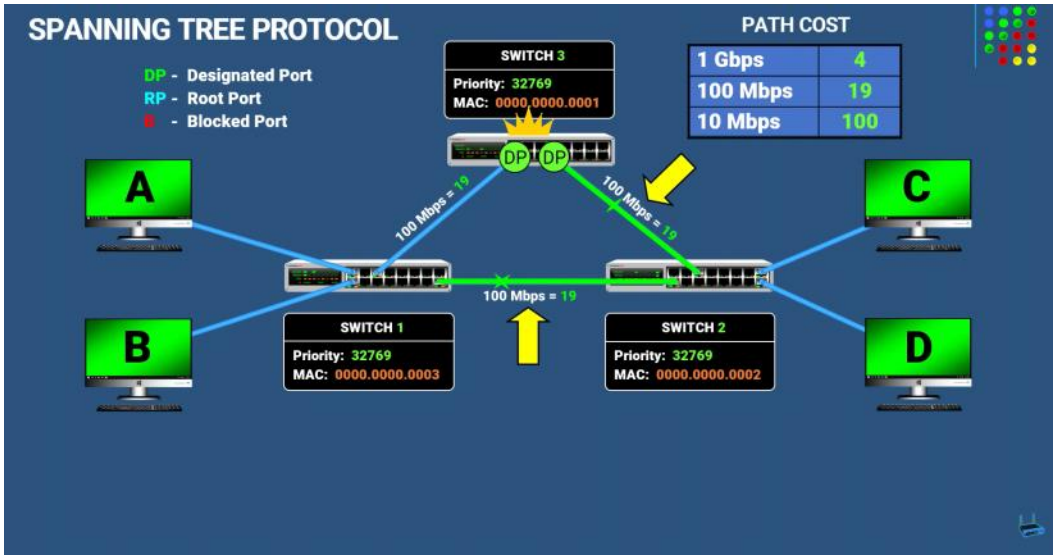
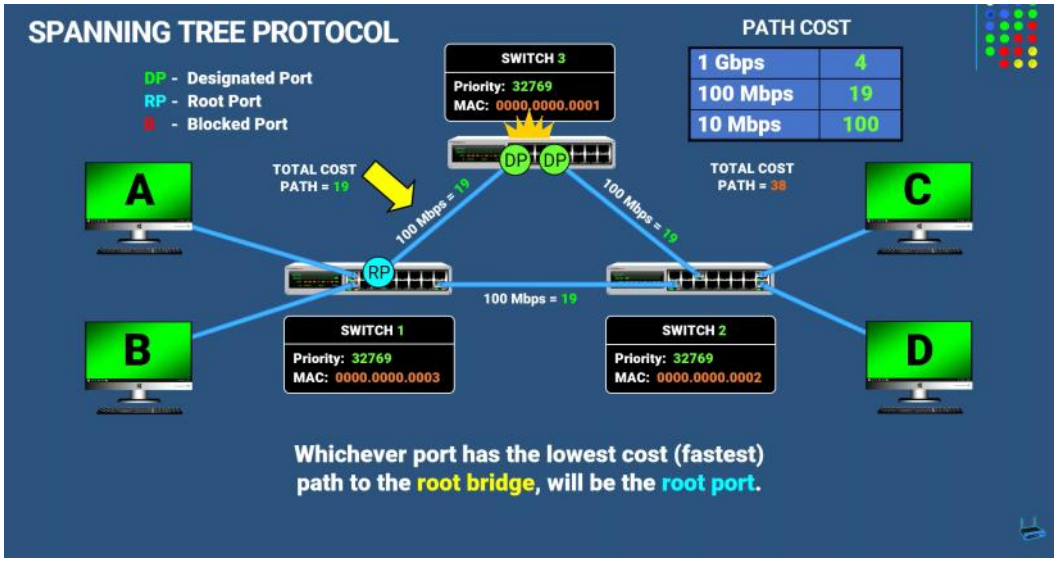
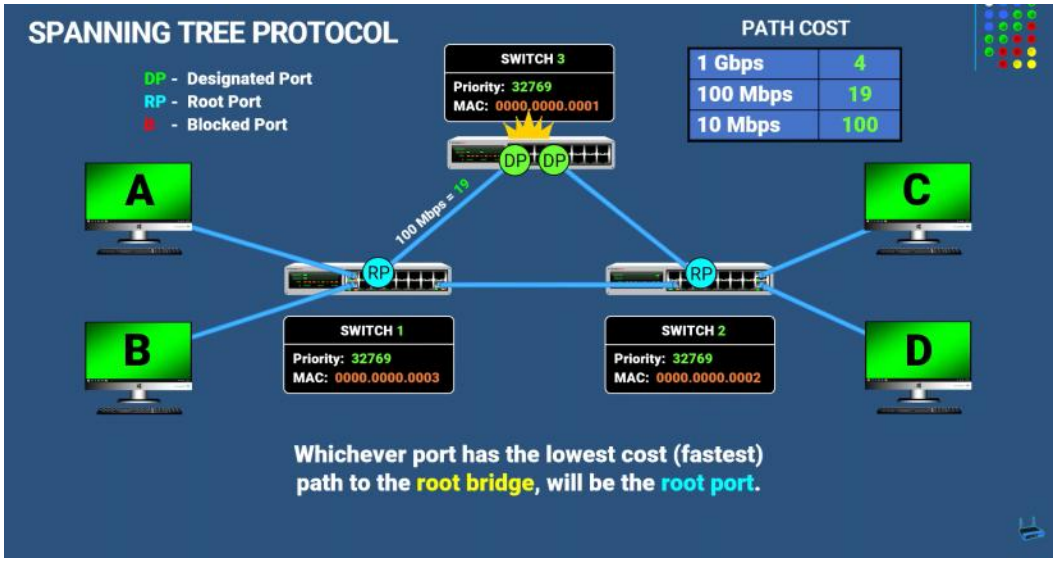
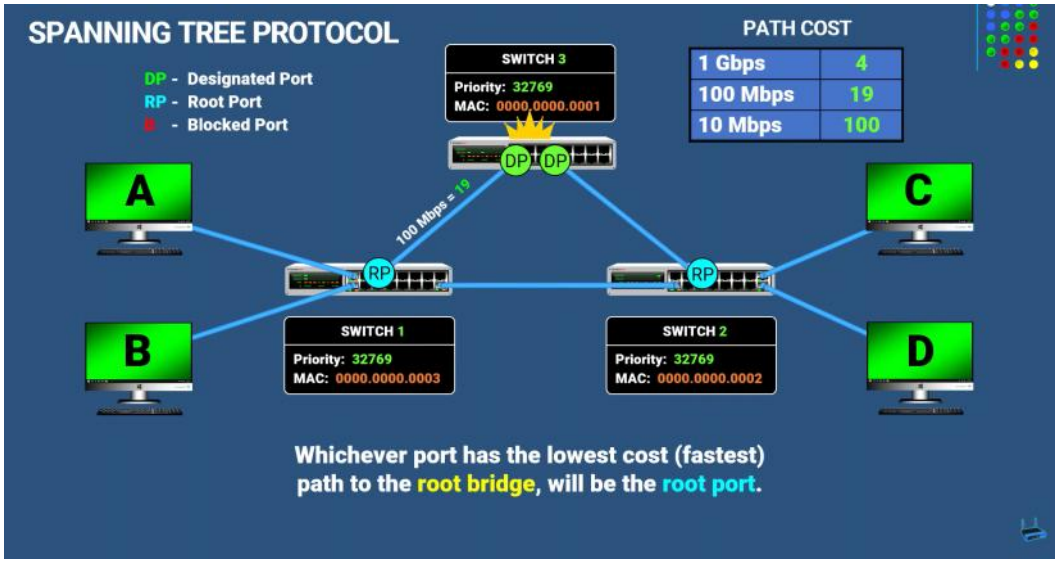






Blocking Ports: Remaining redundant ports are blocked to break potential loops, though they can be reactivated automatically if a link failure occurs.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RP - Root Port
B - Blocked Port
PATH COST
1 Gbps 4
100 Mbps 19
10 Mbps 100
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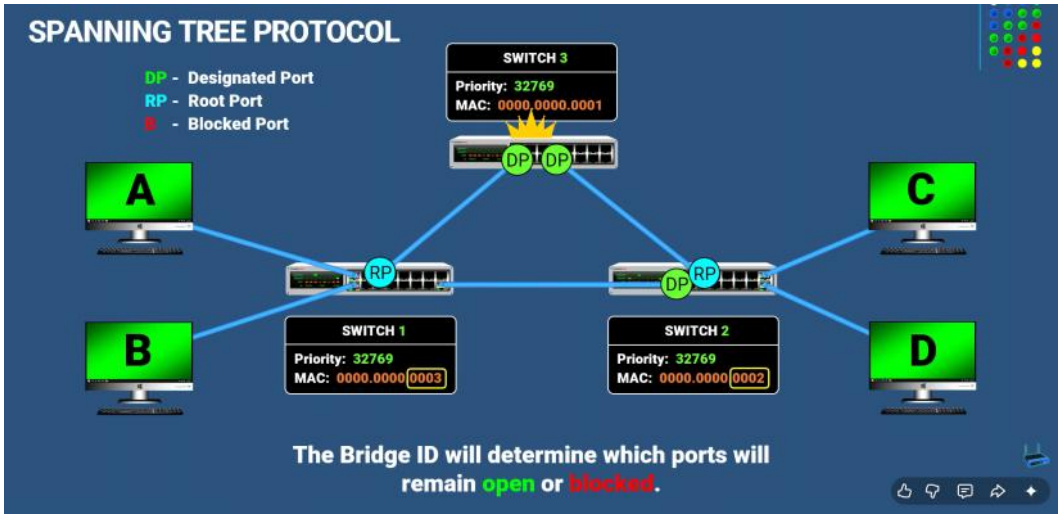
MAC: 0000.0000.0001
SWITCH 1
Priority: 32769
MAC: 0000.0000.0003
SWITCH 2
Priority: 32769
MAC: 0000.0000.0002
100 Mbps = 19
A
B
C
D
Whichever port has the lowest cost (fastest) path to the root bridge, will be the root port.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B - Blocked Port
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MAC: 0000.0000.0001
SWITCH 1
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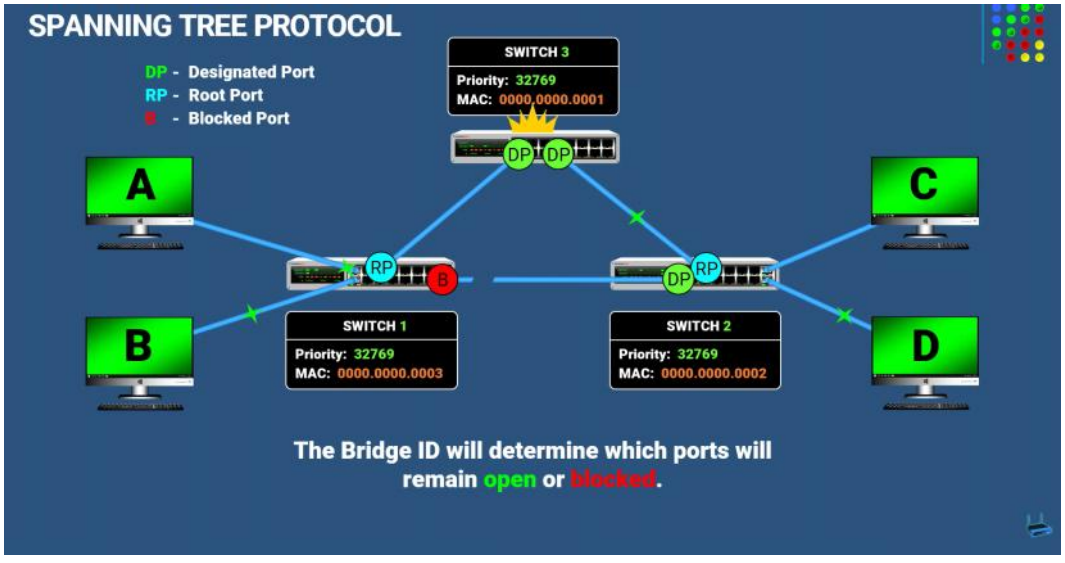
A
B
C
D
The Bridge ID will determine which ports will remain open or blocked.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A
B
C
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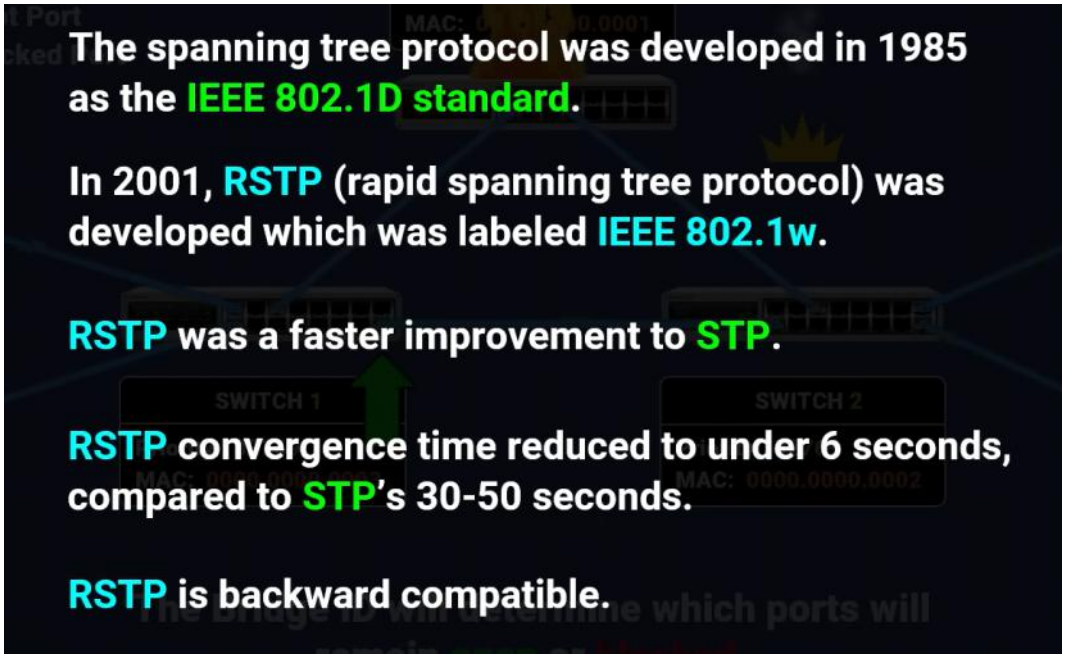
Versions:



- STP (IEEE 802.1d): The original standard, which takes 30 to 50 seconds to adapt to network changes. Developed in 1985.
- RSTP (IEEE 802.1w): A modern, backward-compatible improvement that can adapt to topology changes in under 6 seconds. Developed in 2001.